Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 154–163

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Machine Learning Approach for Predicting the Net Asset Value (NAV) of Mutual Funds based on Portfolio Holdings

Rohit Kumar Sachan[a], Shabanam Kumari[b], Vipul Khandelwal[c], Tarun Kumar[c]*

[a]*SCSET, Bennett University, Greater Noida and 208010, INDIA*
[b]*CED, GL Bajaj Institute of Technology and Management, Greater Noida and 208010, INDIA*
cCSED, Central University of Rajasthan, Ajmer and 305817, INDIA
ertarun123@gmail.com

Abstract

Mutual funds, a cornerstone of modern investment portfolios, offer investors a diversified and professionally managed approach to financial growth. These investment vehicles collect small amount from multiple investors and invests the collective investment further in diversified portfolio of stocks, bonds, or other securities. Mutual funds are characterized by their net asset value (NAV), which represents the per-share value of the fund's assets minus its liabilities. Investors benefit from the expertise of seasoned fund managers who make investment decisions based on their analysis of market trends and company performance. By providing an accessible and diversified investment option, mutual funds have become a popular choice for both novice and seasoned investors seeking to achieve their financial goals. Apart from that, the role of AI is increasing exponentially in various areas. Such AI based tools and algorithms are utilized by various MFs expert to identify the suitable investment portfolios. This paper proposes a technique to analyse the performance of the different stock's portfolios based on the past data. Combining the isolated analysis of different stocks portfolios may be utilized to predict the net asset value prediction (NAV) of MFs. The experiments carried out in this research work uses linear regression analysis to analyse the stock's performance at sub level of the MFs. A case study for the Axis Bluechip fund's is also carried out by using the proposed ML approach. The Yahoo Finance dataset is used for the case study. Previously, machine learning algorithms like linear regression (LR), decision tree regression (DTR), and multivariate regression (MR) are taken to predict and analyse the dataset. These algorithms calculate the NAV of the mutual fund with good accuracy and analyse the best result for the test dataset. The various models are being created from different portfolios to analyse the NAV of the mutual funds. This work proposed hierarchical method to perform the linear regression analysis for the NAV predicting. This work uses R-Square (*R2*) statistical measures to determine the goodness of models. The

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.205

obtained *R2* is 0.86, implying that the created models have good accuracy and performance on the dataset.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

*Keywords:*Decision tree; Machine learning; Mutual funds; Portfolio; Regression; Stocks.

1. Introduction

In the current monetary world, most people are investing or planning for investment as future financial security. There are various investment ways, such as bank deposits, gold, real estate, and stocks. Everyone invests according to their capacity, easy access, and ensuring maximum return with minimum risk. Nowadays, stock investments are rapidly increasing in mutual funds (MFs). Due to its easy access and flexibility, many financial and investment decisions can be made over mutual funds. According to a report [1], the investment in MF crossed Rs. 47 trillion in September 2023 in India. It shows the rapid growth in the MF's investment and the MF's customer base. There is a vast amount of fund holding in the mutual fund, which can be used to classify the MFs. We can also predict and analyze net asset values (NAVs) based on the fund holdings [2]. A mutual fund is a fund in which various investors' investments are collectively invested into various securities, such as stocks, short-term debts, and bonds, by a facilitating organization. The record of the investment is classified as a mutual fund's portfolio. These portfolios are typically classified into four categories. These are *stock or equity funds, money market funds, bond or fixed-income funds, and hybrid funds* [3].

In a *stock or equity portfolio*, the dedicated investment shares are invested into stocks or equity of various sectors listed in the stock exchange market. The NAV of any MF is proportional to the current value of these portfolios. The risk factor of the MF is also directly proportional to these portfolios as the majority share investment is usually invested into these. *Money market portfolios* have comparatively lower risk than other portfolios. Investment into these portfolios is carried out as debt investments into various organizations. Various agencies reevaluate these investments. *Bond or fixed-income portfolios* have a comparatively higher risk than money market portfolios. These are the types of portfolios in which the investor gets paid a fixed over-invested amount. Corporations and the government issue these funds to raise money to expand their business or any project. *Hybrid funds* have an intermediate risk level over the investment.

The NAV of mutual funds mainly depends on the stock's portfolio [3]. However, the probability of the risk and returns are very high due to the dynamic nature of the stock prices. Prediction of the NAV of MFs is challenging for experts. This work proposes ***an approach for the prediction of the NAV of any mutual funds with the use of multiple dependent machine learning (ML) models***. This multi-ML model approach may improve the prediction accuracy of the NAV of MFs compared to past works. The approach presented in this work is also applied to a specific mutual fund as a case study. The major investment of any MF is invested in the stock portfolio. So, analyzing the stock prices is essential for any experts to manage the investments of MFs. The intent behind this work is to propose ***an ML-based analysis approach for better prediction of NAV as mutual funds***. This prediction will help experts identify profitable and low-risk portfolio portfolios for the mutual funds.

There have been several problems with analyzing an MF's NAV and various approaches to analyzing the mutual fund. However, mutual fund analysis and prediction are based on forecasting NAV with time series models and techniques [4]. First, their prediction is based on net asset value and provides no information about internal portfolio holding. Second, prediction and analysis of net asset value using a portfolio have not been done so far.

The primary purpose of this work is to propose an approach for predicting the NAV of any MFs using ML models. In short, the main contributions of our work are to discuss past related works in the domain of the prediction of the NAV of mutual funds, elaborate the proposed ML-based approach to predict the NAV of mutual funds with the help of a case study of Axis Bluechip funds and analyze the proposed modal based on identified accuracy, i.e., 86.62%.

2. Related Work

Since the last two decades, there has been impressive research done and, as yet, going on the mutual fund (MF) and NAV prediction or forecasting. The analysis of mutual funds turned out to be a piece of research towards the start of the 20th century. This proposed work analyzed the internal holding problems of the mutual fund portfolios. This proposed work explores the ML techniques because these techniques exploit characteristics of the portfolio holdings to predict the NAV of mutual funds. It unveils the interactions in the relation between portfolio characteristics and the future performance of MFs.

In [5], authors use various time series (TS) models to forecast the NAV of MFs. This work also compares the various models like ARIMA (autoregressive integrated moving average), DT (decision tree), GBT (gradient boosted trees), and RF (random forest) for predicting gold price funds. They use 90-10 rules for data decomposition of training and testing data and measure the performance using RMSE (root mean square error). They find that RF outperforms DT, GBT, and ARIMA in predicting the future gold price. In another work [6], authors apply ML algorithms, namely the Gaussian hybrid clustering algorithm, to improve the accuracy and timeliness of fund classification. They have also compared the classic K-means, network clustering, and proposed GMM method. They find a higher probability by the GMM method for fund classification. In [7], the authors propose an ML-based approach to predict the trends of stock prices. The approach uses historical prices of stocks along with sentiment data and predicts future trends. The proposed approach achieves 70% accuracy in predicting daily stock prices. Another work [8] proposes an LSTM (long-sort time term memory)-based approach for stock price forecasting as the price trends of the stock flow with TS like LSTM model. The model is tested over the Google and Nike stock data for prediction. The prediction has been made in different epochs to find the most relevant result with minimum loss value. The epoch models are created for 12, 25, 50, and 100. After getting the results from each epoch, the authors compare the values and find out the best epoch model for predicting future price trends for each stock. Similarly, work [9] proposes an ML-based approach using DT and the XGboost model. The work analyses the equity funds factors of MFs. The proposed approach achieves 80% accuracy in predicting the NAV of the MFs.

In [10], authors use FFLANN (feedback functional link artificial neural network) to predict NAV. They use the Birla Sun Life Equity fund, ICICI Pro top100, SBI Bluechip fund, and UTI Equity fund dataset for evaluation. They calculate the model's accuracy based on the RMSE and MAPE (mean absolute percentage error). In another paper [11], authors use four different predictive models for the stock from three sectors: automobile, IT sector, and banking sector. They use DL, RNN (recurrent neural network), CNN (convolutional Neural Network), MLP (multilayer perceptron), and LSTM approach as a prediction model. Out of these, DL outperforms the best. CNN performs best for automobile sector stock, i.e., TATA motors, and LSTM performs best for the banking sector, i.e., Bank of America. Work [12] creates a TS model for forecasting NAV for mutual fund asset allocation. This work uses a Vanguard Wellington Fund dataset and various models like the linear trend, quadratic trend, and exponential model. Another work [13] uses the LSTM approach to predict the value of Nifty50 from NSE. This author uses five years of dataset, divided into two parts, mainly for validation and training. A five-stage methodology is used in this work. These stages are raw data, data processing, feature extraction, training neural network, and output generation. All these works are summarized in Table 1 in terms of type of problem, proposed model, compared model, used dataset/funds, and reported performance/ or used performance metrics. All these works mainly forecast the NAV of MFs based on historical data or time series data analysis and do not consider the internal holdings of MFs, which may also play a significant role in the NAV prediction of MFs.

3. Proposed Approach

This work proposes an analytical approach for predicting the NAV of mutual funds, which depends on the investment portfolios and shared investment of MFs. In general trends, most MFs are invested in stock-based portfolios. This approach analyzes the MFs' significant stock holdings and stock portfolios. In this, an individual ML model is used for each stock portfolio, and it predicts the future stock trends of each portfolio based on their historical price. Later, a DT model combines the obtained prediction from the individual ML model of each stock price, which will be used to predict the future NAV of the MFs. The proposed approach can be utilized for better

investment plans for the risk assumed. This work uses a case study for the validation purpose of the proposal. It is presented in Section 3.1, and the workflow of the proposed approach is also discussed in Section 3.2.

Table 1: Summery of related work.

Ref.	Problem	Proposed model	Compared models	Dataset/Funds	Performance
[5]	Prediction	Time series-based	ARIMA, DT, GBT, RF	Gold price funds	RMSE
[6]	Classification	GMM	K-means, network clustering	-	-
[7]	Prediction	ML-based	-	Historical stock data	Acc= 70%
[8]	Prediction	LSTM-based	Different epoch models	Google and Nike stock data	-
[9]	Prediction	DT with XGboost	-	Equity funds	Acc= 80%
[10]	Prediction	FFLANN	-	Birla Sun Life Equity, ICICI Pro top100, SBI Bluechip, UTI Equity	RMSE, MAPE
[11]	Prediction	DL-based	RNN, CNN, MLP, LSTM	Automobile, IT, Banking sector	Acc
[12]	Forecasting	Time series-based	Linear, quadratic, exponential model	Vanguard Wellington fund	-
[13]	Prediction	LSTM	-	Nifty50	-

3.1. Case Study

This work uses case study data to train the model to verify and test the proposed hypothesis. This case study focuses on finding the internal fund holding of the Axis Bluechip fund that will help us provide the prediction values of the stocks that come under the fund. Stocks like Infosys Ltd, Bajaj Finance Ltd, ICICI Bank Ltd, HDFC Bank Ltd, Tata Consultancy Services Ltd and DMART (Avenue Supermarts Ltd.) hold more than 50% of the fund portfolio holding of Axis Bluechip fund [14] (c.f. Table 2).

Table 2: Details of portfolio holding of Axis Bluechip fund.

Stock Invested In	Sectors	Percentage of fund holding
Infosys Ltd.	Software	9.77%
Bajaj Finance Ltd.	NBFC	9.69%
ICICI Bank Ltd.	Bank	9.09%
HDFC Bank Ltd.	Bank	7.46%
TCS Ltd.	Software	7.00%
DMART (Avenue Supermarts Ltd.)	Retailing	6.97%

Axis Bluechip fund dataset is used due to its highly equity holding portfolio fund, which has invested portfolio in sectors like banking, IT, NBFC, and retailing sectors from Yahoo Finance [15]. The dataset contains the *stock symbol, stock series, stock date and previous closing, opening, high, low, final, closing, average prices, and total traded volume*. This work extracts the day-by-day closing price and the date of each portfolio from the dataset. Day-by-day stock prices are preferable for investors' decision-making process in which stocks will be purchased or sold based on the market's closing price. This work also considers the NAV of a particular fund in which its portfolios hold all the stocks.

3.2. Workflow

This work predicts the NAV of MFs based on the equity percentage of their stock portfolio. The stock portfolio is selected from the internal holding of the MF. There can be an 'n' number of the stock portfolio in the MF, which can be based on the equity shareholding. During training, our model is trained using LR to analyze the stock price trends, and then a DTR will be used to predict the closing price of the stock portfolio. After the training, our model will be ready for testing. Here, both LR and DTR are used to check and analyze portfolio performance. As the

closing stock portfolio price is predicted, a new MR model is created to analyze the NAV of mutual funds. The MR imports the predicted portfolio dataset into the model. For every date, a predicted and analyzed mutual fund's NAV is taken out from the model. Ultimately, this work tests the model using R-Square (*R2*) statistical measure (c.f. Figure 1).

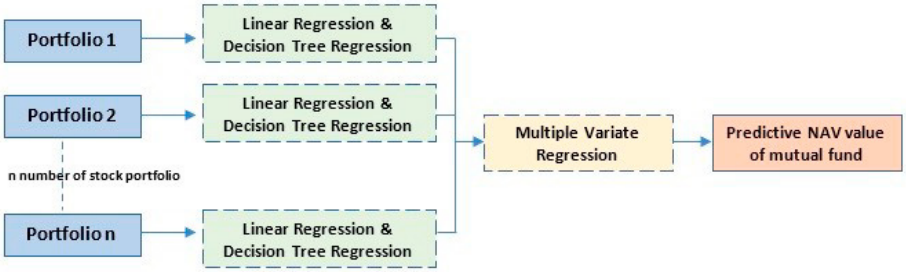


Fig. 1: Flow chart of the working models to analyze and predictive the NAV value of mutual fund.

4. Result and Analysis

This section presents the details about the dataset, the different data processing models this work has used, and the obtained result with the analysis.

4.1. Dataset Overview

While analyzing a mutual fund's portfolio, the investor can analyze the fund's future prediction value so that it is a good idea about the return on investment. Here, this work builds a common dataset using big data techniques in which we merge the data of six portfolios from the six different datasets. These techniques will help us to manage or organize the data. This work considers the data range from 2018-01-01 to 2021-12-31, and all the datasets come from the different locations by which we can create models on the dataset. While we create models, the datasets are divided into the 80-20 ratio. 80% of our data will be used for training the models, and 20% will be used for testing purposes. In this dataset, all portfolio-holding stock values belong to an independent variable, and NAV comes under the dependent variable. The final ready-to-us dataset of the Axis Bluechip fund is sniped in Table 3.

Table 3: Final dataset ready to create models for analyzing the NAV of the Axis Bluechip fund with its internal portfolio holding.

S.No.	Date	INFY	BAJ	DMART	HDFC	ICICI	TCS	NAV
1	02-01-2018	514.85	1724.65	1160	936.175	309.7	1315.6	17.78
2	03-01-2018	510.65	1715.3	1184.55	926.325	315	1319.325	17.79
3	04-01-2018	507.7	1754	1203.1	929.95	314.7	1328.55	17.83
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
981	29-12-2021	1885.55	6911.65	4683.95	1453.85	735.7	3694.7	23.82
982	30-12-2021	1892.85	6871.1	4639.35	1461.5	735.7	3733.75	23.79
983	31-12-2021	1887.75	6977.3	4671.45	1479.4	740.15	3738.35	24.01

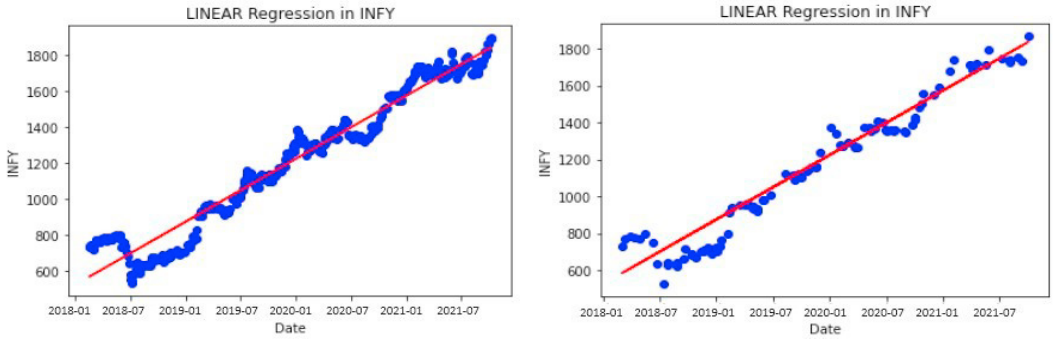
4.2. Data Prepossessing Models

The ML models use the training and testing dataset of the Axis Bluechip fund in the portfolios, which gives a brief idea about the regression models of the MF. This work uses techniques such as imputation, scaling, normalization, and standardization for data preparation and featurization. These techniques improve the efficiency of ML models. Here, while creating the model, the dataset is divided into two parts. Where 20% of the dataset is used for testing, and 80% is used for creating the regression models over portfolio prices and date range. In the pre-

processing data model, this work has mainly used LR and DTR models to analyze the portfolio's price value. Here, this paper creates six different models based on the LR and DTR, using the past dataset of the stock portfolio, by which the future price of the stock is forecasted to train the previous stock price dataset. While creating the model, this work will get the original price, the predicted forecast price, and the valid price, the actual closing price of the respective portfolio.

Model 1 - Stock of Infosys for LR and DTR

In Figure 2a, the LR model is used to analyze the past trend and volatility of the portfolio stock Infosys, which holds 9.77% of the portfolio of the invested amount. Whereas in Figure 2b, the model is tested on this portfolio by selecting 20% of the data for training is from the raw dataset. In Figure 4a, the DTR model is applied to predict the closing price value of the portfolio stock Infosys. In this model, the last 100 days show the actual closing price and predicted closed price values of the Infosys portfolio.

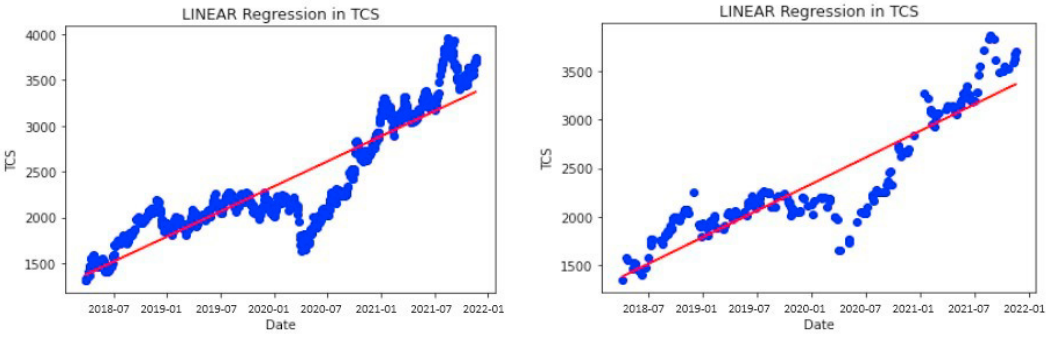


(a) Linear regression over closing price of Infosys (b) Testing of Linear regression model over closing price of Infosys

Fig. 2: Infosys portfolio

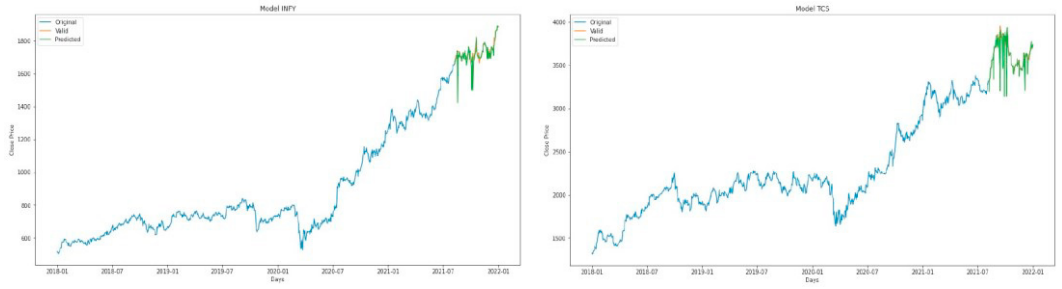
Model 2 - Stock of TCS for LR and DTR

In Figure 3a, the LR model is used to analyze the past trend and volatility of the portfolio stock TCS, which holds 7.00% of the portfolio of the invested amount. Whereas in Figure 3b, the model is tested on this portfolio by selecting 20% of the data for training from the raw dataset. In Figure 4b, the DTR model is applied to predict the closing price value of the portfolio stock TCS. In this model, the last 100 days show the actual closing price and predicted closed price values of the TCS portfolio.



(a) Linear regression over closing price of TCS (b) Testing of Linear regression model over closing price of TCS

Fig. 3: TCS portfolio

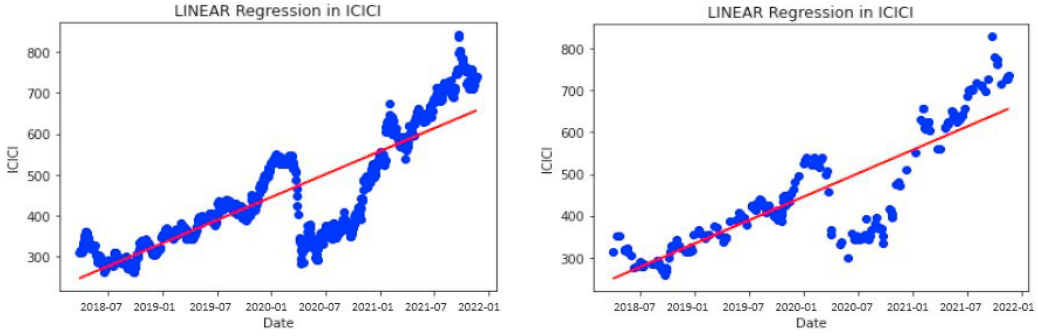


(a) Infosys (b) TCS

Fig. 4: Decision tree regression - Infosys and TCS

Model 3 - Stock of ICICI Bank for LR and DTR

In Figure 5a, the LR model is used to analyze the past trend and volatility of the portfolio stock ICICI Bank, which holds 9.09% of the portfolio of the invested amount. Whereas in Figure 5b, the model is tested on this portfolio by selecting 20% of the data for training from the raw dataset. In Figure 7a, the DTR model is applied to predict the closing price value of the portfolio stock ICICI Bank. In this model, the last 100 days show the actual closing price and predicted closed price values of the ICICI BANK portfolio.



(a) Linear regression over closing price of ICICI (b) Testing of Linear regression model over closing price of ICICI

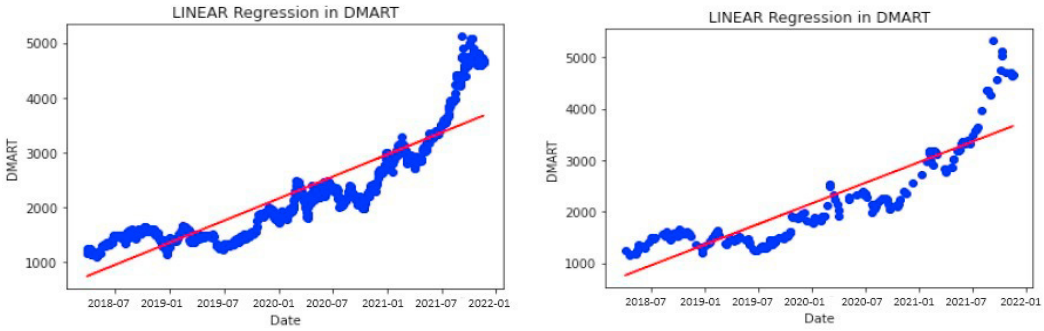
Fig. 5: ICICI portfolio

Model 4 - Stock of DMART for LR and DTR

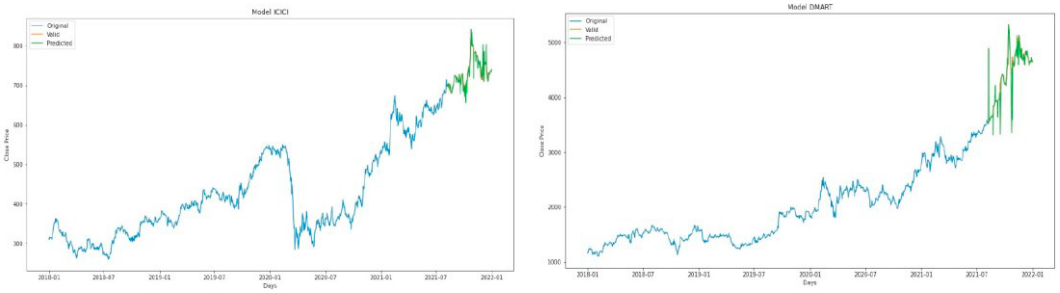
In Figure 6a, the LR model is used to analyze the past trend and volatility of the portfolio stock Avenue Supermart, which holds 6.97% of the portfolio of the invested amount. Whereas in Figure 6b, the model is tested on this portfolio by selecting 20% of the data for training from the raw dataset. In Figure 7b, the DTR model is applied to predict the closing price value of the portfolio stock Avenue Supermart. In this model, the last 100 days show the actual closing price and predicted closed price values of the DMART Supermart portfolio.

Model 5 - Stock of Bajaj Finance for LR and DTR

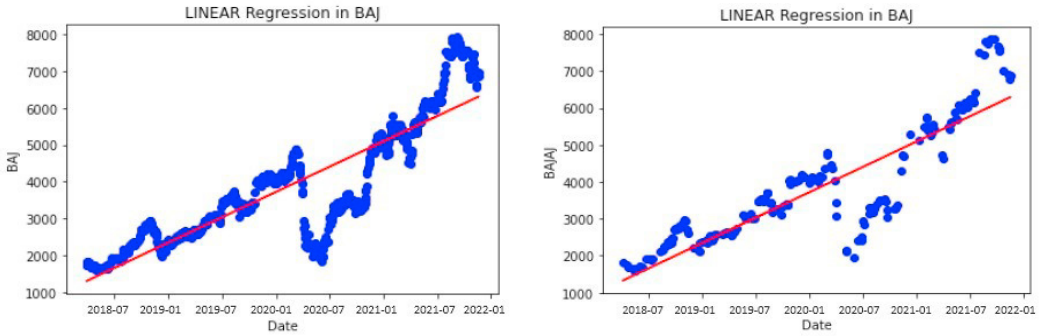
In Figure 8a, the LR model is used to analyze the past trend and volatility of the portfolio stock Bajaj Finance, which holds 9.69% of the portfolio of the invested amount. Whereas in Figure 8b, the model is tested on this portfolio by selecting 20% of the data for training from the raw dataset. In Figure 10a, the DTR model is applied to predict the closing price value of the portfolio stock Bajaj Finance. In this model, the last 100 days show the actual closing price and predicted closed price values of the Bajaj Finance portfolio.



(a) Linear regression over closing price of DMART (b) Testing of Linear regression model over closing price of DMART
Fig. 6: DMART portfolio



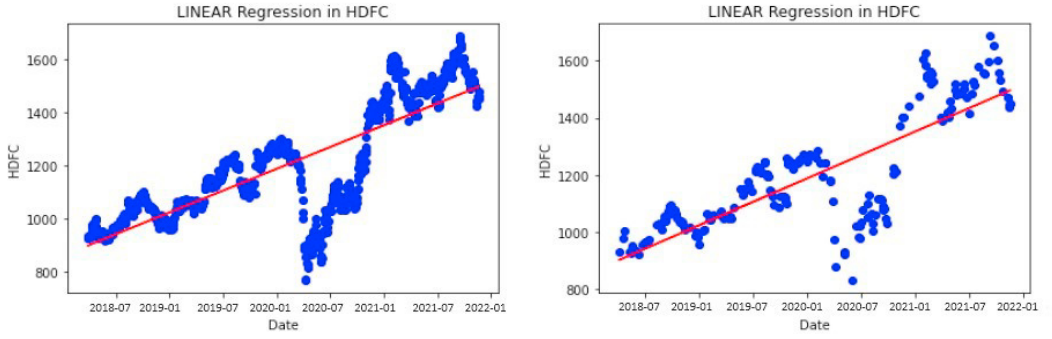
(a) ICICI (b) DMART
Fig. 7: Decision tree regression - ICICI and DMART



(a) Linear regression over closing price of BAJAJ (b) Testing of Linear regression model over closing price of BAJAJ
Fig. 8: BAJAJ portfolio

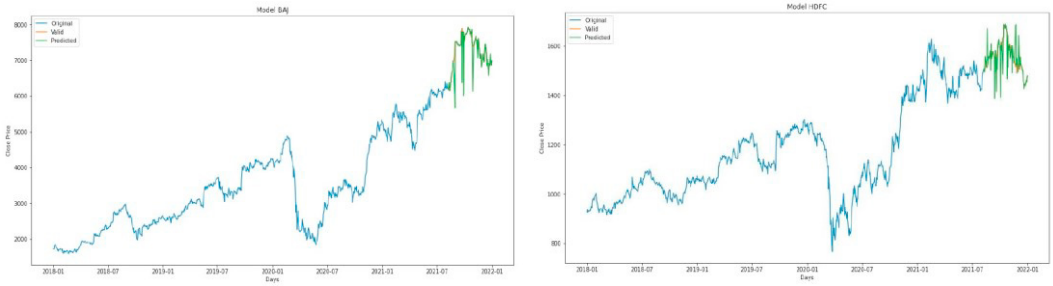
Model 6 - Stock of HDFC Bank for LR and DTR

In Figure 9a, the LR model is used to analyze the past trend and volatility of the portfolio stock HDFC Bank, which holds 7.46% of the portfolio of the invested amount. Whereas in Figure 9b, the model is tested on this portfolio by selecting 20% of the data for training from the raw dataset. In Figure 10b, the DTR model is applied to predict the closing price value of the portfolio stock HDFC Bank. In this model, the last 100 days show the actual closing price and predicted closed price values of the HDFC BANK portfolio.



(a) Linear regression over closing price of HDFC (b) Testing of Linear regression model over closing price of HDFC

Fig. 9: HDFC portfolio



(a) Bajaj Finance (b) HDFC

Fig. 10: Decision tree regression - Bajaj Finance and HDFC

4.3. Multivariate LR Table For NAV

In the MR, portfolio-holding stocks are the independent variable, whereas NAV is the dependent variable. Table 4 shows the result in which random is selected to analyze the predicted NAV and actual NAV. *R2*, or the coefficient of determination [16], is taken to analyze the model. In Table 4, all values of *R2* are the same, representing the consistency and stability of the proposed work.

Table 4: Comparative Results using multivariate regression to predict NAV from portfolio holding.

Date	Actual NAV	Predicted NAV	*R2* (coefficient of determination)
12-12-2018	16.98	16.53	0.8662262206661
29-06-2018	17.12	17.98	0.8662262206661
12-11-2018	16.20	15.93	0.8662262206661
13-03-2019	17.64	18.03	0.8662262206661
23-09-2019	18.36	17.90	0.8662262206661
01-01-2020	18.87	17.60	0.8662262206661
15-06-2020	14.72	15.03	0.8662262206661
16-06-2021	21.53	22.12	0.8662262206661
31-12-2021	24.01	23.17	0.8662262206661

4.4. . Results

This work achieves a 0.8662262206661 coefficient of determination, representing that our dataset fits with the accuracy of 86.62% for the model. Comparative result between actual NAV and predictive NAV is shown in figure 11.

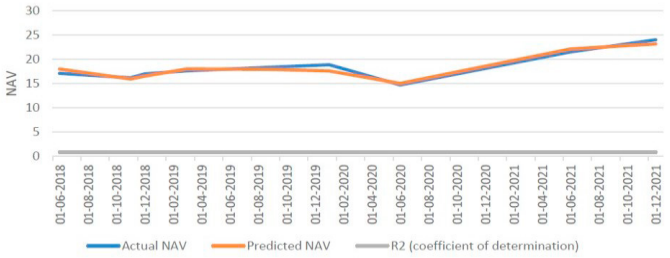


Fig. 11: Comparative result between actual NAV and predictive NAV.

5. Conclusion and Future Scope

Creating models of LR and DTR is a notable and critical complex optimization to predict portfolio value. This work presents the benefits of analyzing the NAV value from internal portfolio holding using MR, for example, accomplishing a good return on investment, analyzing the investment risk, and expanding the accuracy with a new approach to finding the NAV of the MF. The testing was done on the portfolio stocks dataset using both LR and DTR, which gives us a good understanding of the stock price volatility. This proposed work is based on NAV control and prediction of MF performance. So, according to the analysis of the MF performance, the agent controlling the portfolio can change for high value on return on investment and make out more profit. In real life, investors may also use this predicted NAV to maximize their profits. There are various models and prediction methods out of which portfolio price can be predicted with more accuracy and can take a large portfolio dataset to improve the result of the portfolio.

References

[1] IIFL Securties."Key Mutual Fund Trends Observed In September 2023."https://www.indiainfoline.com/article/general-blog/ key-mutual-fund-trends-observed-in-september--1697615740380_1.html. [Online; accessed 01-November-2023].

[2] Yılmaz, Ümit, and ÂliYurdunOrbak. "Prediction of Turkish mutual funds' net asset value using the fund portfolio distribution." *Neural Computing and Applications* (2023): 1-18.

[3] Hayes, Adam. 2023. "Mutual Funds: Different Types and How They Are Priced."https://www.investopedia.com/terms/m/mutualfund. asp. [Online; accessed 10-April-2023].+

[4] Kaniel, Ron, Zihan Lin, Markus Pelger, and Stijn Van Nieuwerburgh. "Machine-learning the skill of mutual fund managers." *Journal of Financial Economics* 150, no. 1 (2023): 94-138.

[5] Rady, E. H. A., Haitham Fawzy, and Amal Mohamed Abdel Fattah. "Time series forecasting using tree-based methods." *J. Stat. Appl. Probab* 10, no. 1 (2021): 229-244.

[6] Chen, Xiaofei, Shujun Ye, and Chao Huang. "Cluster-Based Mutual Fund Classification and Price Prediction Using Machine Learning for Robo-Advisors." *Computational Intelligence and Neuroscience* 2021 (2021).

[7] Nayak, Aparna, MM Manohara Pai, and Radhika M. Pai. "Prediction models for Indian stock market." *Procedia Computer Science* 89 (2016): 441-449.

[8] Moghar, Adil, and Mhamed Hamiche. "Stock market prediction using LSTM recurrent neural network." *Procedia Computer Science* 170 (2020): 1168-1173.

[9] Promptsook, Nuttrachai, and KitsanaWaiyamai. "Thai Equity Mutual Fund net asset value return prediction using internal factors." In *2021 2nd International conference on big data analytics and practices (IBDAP)*, pp. 119-123. IEEE, 2021.

[10] Anish, C. M., and Babita Majhi. "Prediction of mutual fund net asset value using low complexity feedback neural network." In *2016 IEEE international conference on current trends in advanced computing (ICCTAC)*, pp. 1-5. IEEE, 2016.

[11] Hiransha, Ma, E. Ab Gopalakrishnan, Vijay Krishna Menon, and K. P. Soman. "NSE stock market prediction using deep-learning models." *Procedia computer science* 132 (2018): 1351-1362.

[12] Lawrence, Kenneth D., Gary Kleinman, and Sheila M. Lawrence. "Time series models to predict the net asset value (nav) of an asset allocation mutual fund vwelx." In *Handbook of Financial Econometrics and Statistics*, pp. 2445-2460. Springer New York, 2015.

[13] Roondiwala, Murtaza, Harshal Patel, and Shraddha Varma. "Predicting stock prices using LSTM." *International Journal of Science and Research (IJSR)* 6, no. 4 (2017): 1754-1756.

[14] Zerodha. "Axis Bluechip Fund."https://coin.zerodha.com/mf/fund/INF846K01DP8. [Online; accessed 10-July-2023].

[15] Yahoo Finance. "Yahoo Finance."https://finance.yahoo.com/. [Online; accessed 10-July-2023].

[16] Byjus. "Coefficient of Determination."https://byjus.com/maths/coefficient-of-determination. [Online; accessed 20-July-2023].